Outsourcing Purchase Prices

May 2026

1 Regression Results

Overview

The dependent variable in all regressions is $\ln p_{fjct}$, the log outsourcing purchase unit price (CNY per unit) paid by firm f for product j in city c in year t (2017). Three groups of explanatory variables are considered.

- **Firm-level variables (x):** $\ln \text{Output}_{ft}$ (log total VAT sales, size proxy); $\ln K_{ft}$ (log total assets); $\ln Q_{fjt}$ (log of the firm's own sales quantity of product j). $\ln \text{Output}$ and $\ln K$ vary at the firm×year level and are absorbed by firm FE; $\ln Q_{fjt}$ varies at the firm×product level and remains identified under firm FE.

- **Market-level variables (z):** $\ln n^B_{jct}$ (log number of buyers of product j in city c); $\ln n^S_{jct}$ (log number of sellers); $\ln q^m_{jct}$ (log total market purchase quantity); $\ln \bar{p}^B_{jct}$ (log buy-side leave-one-out market average price); $\ln \bar{p}^S_{jct}$ (log sell-side leave-one-out market average price). Both leave-one-out prices subtract the firm's own buy/sell transactions from the city–product totals, removing the mechanical correlation between the dependent variable and these controls.

- **Product-similarity variables (s):** S_{mj} (input_similarity); C_{mj} (output_similarity). Both vary at the firm×product level.

Sample. The sample consists of 3,410 firms randomly drawn from the VAT database. The sample passes through the following filtering steps:

- **Buy-side filter**: 3,376 of the 3,410 firms have at least one valid 9-digit product code in the purchase records.
- **Outsourcing definition**: an outsourcing product requires the same firm to appear on both the buy and sell side for the same product code. Only 2,108 firms satisfy this condition.

- **Intermediary removal**: 233 firms with outsourcing ratio > 90% are dropped, leaving 1,875 firms in the final regression panel.

Product portfolio. Among the 3,376 buy-side firms, each purchases on average approximately 125 distinct product types. In the final regression panel 1,875 firms, the mean is 25.0 outsourcing products per firm (median 3), spanning 2,143 distinct product types and 262 cities.

Fixed effects and identification. All FE specifications include firm (δ_f), product (δ_j), and/or city (δ_c) fixed effects; standard errors are clustered at the firm level. Because the data cover a single cross-section (2017), the firm-level variables $\ln \text{Output}$ and $\ln K$ are perfectly collinear with firm FE and are identified only in OLS specifications.

1.1 T1: Firm-Level Variables

The baseline equation is

$$\ln p_{fjct} = \gamma_1 \ln \text{Output}_{ft} + \gamma_2 \ln K_{ft} + \gamma_3 \ln Q_{fjt} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \tag{1}$$

Table 1: Variable Definitions — Table T1

Variable	Definition
p_{fjct}	Log outsourcing purchase unit price, firm f, product j, city c, year t.
$\ln \text{Output}_{ft}$	Log annual VAT sales
$\ln K_{ft}$	Log total assets
$\ln Q_{fjt}$	Log of the firm's own output quantity of product j
δ_f, δ_j, δ_c	Firm, product, and city fixed effects.

Key findings.

- In OLS, firm size $\ln \text{Output}$ is significantly positive ($+0.234^{***}$): larger firms pay higher outsourcing prices (maybe better quality?); capital $\ln K$ is significantly negative (-0.156^{***}): more capital-rich firms pay lower prices.
- $\ln Q$ is significantly negative in both OLS and the firm-FE column (-0.177^{***}, -0.169^{***}): larger production quantities go with lower outsourcing prices, consistent with quantity discounts.

Table 2: T1: Firm-Level Variables Only

	(1) OLS	(2) +Firm FE
$\ln \text{Output}$	0.234^{***}	—
	(0.051)	
$\ln K$	-0.156^{***}	—
	(0.047)	
$\ln Q$	-0.177^{***}	-0.169^{***}
	(0.013)	(0.013)
Cons	3.527^{***}	4.969^{***}
	(0.452)	(0.081)
Firm FE	No	Yes
Product FE	No	No
City FE	No	No
N	29,247	28,915
Adj. R^2	0.081	0.311

1.2 T2: Firm-Level Variables and Product Similarity

The equation adds product-similarity terms:

$$\ln p_{fjct} = \gamma_1 \ln \text{Output}_{ft} + \gamma_2 \ln K_{ft} + \gamma_3 \ln Q_{fjt} + \beta_1 S_{mj} + \beta_2 C_{mj} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \quad (2)$$

Table 3: Variable Definitions — Table T2

Variable	Definition
S_{mj}	Input similarity.
C_{mj}	Output similarity
m	Core product: with the highest net production.

Key findings.

- In OLS, firm size is positive ($+0.267^{***}$) and capital is negative (-0.112^{***}), as in T1.
- Similarity S_{mj}, C_{mj} are strongly positive in OLS ($+1.491^{***}$, $+1.481^{***}$) and attenuate sharply under Product FE to marginal significance (0.186^{*}, 0.263^{*}): the similarity effect is largely explained by product-level heterogeneity.
- $\ln Q$ is significantly negative from OLS (-0.226^{***}) to full FE (-0.070^{***}).

Table 4: T2: Firm-Level Variables and Product Similarity

	(1) OLS	(2) +Firm FE	(3) +Firm+Prod	(4) +Firm+Prod+City
$\ln \text{Output}$	0.267^{***}	—	—	—
	(0.042)			
$\ln K$	-0.112^{***}	—	—	—
	(0.040)			
$\ln Q$	-0.226^{***}	-0.193^{***}	-0.070^{***}	-0.070^{***}
	(0.013)	(0.014)	(0.008)	(0.008)
S_{mj}	1.491^{***}	0.460^{**}	0.186^{*}	0.186^{*}
	(0.255)	(0.186)	(0.109)	(0.109)
C_{mj}	1.481^{***}	1.361^{***}	0.263^{*}	0.263^{*}
	(0.285)	(0.210)	(0.141)	(0.141)
Cons	1.826^{***}	4.846^{***}	4.274^{***}	4.274^{***}
	(0.452)	(0.076)	(0.047)	(0.048)
Firm FE	No	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
City FE	No	No	No	Yes
N	29,237	28,905	28,552	28,552
Adj. R^2	0.127	0.321	0.595	0.592

1.3 T3: Market-Level Variables Only

The market-level specification is

$$\ln p_{fjct} = \lambda_1 \ln n^B_{jct} + \lambda_2 \ln n^S_{jct} + \lambda_3 \ln q^m_{jct} + \lambda_4 \ln \bar{p}^B_{jct} + \lambda_5 \ln \bar{p}^S_{jct} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \quad (3)$$

Table 5: Variable Definitions — Table T3

Variable	Definition
$\ln n^B_{jct}$	Log number of buyers of product j in city c
$\ln n^S_{jct}$	Log number of sellers of product j in city c
$\ln q^m_{jct}$	Log total market purchase quantity of product j in city c
$\ln \bar{p}^B_{jct}$	Log buy-side leave-one-out market average price
$\ln \bar{p}^S_{jct}$	Log sell-side leave-one-out market average price

Key findings.

- The contrast between the two leave-one-out prices is the key new finding. The buy-side price $\ln \bar{p}^B$ is strongly positive in OLS/Firm FE (0.440^{***}, 0.365^{***}) but collapses to -0.029^{*} under Product FE—the earlier strong correlation was driven by product composition and mechanical correlation.
- The sell-side price $\ln \bar{p}^S$, by contrast, remains positive and significant even under full FE (0.052^{***}), carrying price information beyond product FE.
- The buyer count $\ln n^B$ is robustly positive (0.115^{***}), and market volume $\ln q^m$ turns negative under Product FE (-0.100^{***}).

Table 6: T3: Market-Level Variables Only

	(1) OLS	(2) +Firm FE	(3) +Firm+Prod	(4) +Firm+Prod+City
$\ln n^B$	0.170^{***}	0.098^{***}	0.115^{***}	0.115^{***}
	(0.051)	(0.029)	(0.040)	(0.040)
$\ln n^S$	0.104^{*}	0.064^{*}	-0.006	-0.006
	(0.061)	(0.036)	(0.035)	(0.035)
$\ln q^m$	-0.009	-0.029^{*}	-0.100^{***}	-0.100^{***}
	(0.017)	(0.015)	(0.017)	(0.017)
$\ln \bar{p}^B$	0.440^{***}	0.365^{***}	-0.029^{*}	-0.029^{*}
	(0.017)	(0.016)	(0.017)	(0.017)
$\ln \bar{p}^S$	0.302^{***}	0.246^{***}	0.052^{***}	0.052^{***}
	(0.020)	(0.016)	(0.011)	(0.011)
Cons	0.218	1.643^{***}	4.912^{***}	4.912^{***}
	(0.259)	(0.208)	(0.302)	(0.302)
Firm FE	No	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
City FE	No	No	No	Yes
N	46,444	45,956	45,669	45,669
Adj. R^2	0.310	0.468	0.604	0.602

1.4 T4: Firm-Level and Market-Level Variables

The combined firm and market specification is

$$\ln p_{fjct} = \boldsymbol{\gamma}'\mathbf{x}_{ft} + \boldsymbol{\lambda}'\mathbf{z}_{jct} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}, \tag{4}$$

where $\mathbf{x}$ includes the production quantity $\ln Q_{fjt}$.

Key findings.

- In OLS, firm size is positive ($+0.157^{***}$) and capital is negative (-0.101^{***}); $\ln Q$ is significantly negative from OLS (-0.126^{***}) to full FE (-0.061^{***}).
- The buy-side price $\ln \bar{p}^B$ is significant in OLS/Firm FE (0.503^{***}, 0.403^{***}) but drops to -0.008 under Product FE, whereas the sell-side price $\ln \bar{p}^S$ stays positive and significant under full FE (0.047^{***}).
- Market volume $\ln q^m$ is positive in OLS (0.077^{***}) but negative under full FE (-0.078^{***}); the buyer count is positive under full FE (0.106^{**}).

Table 7: T4: Firm-Level and Market-Level Variables

	(1) OLS	(2) +Firm FE	(3) +Firm+Prod	(4) +Firm+Prod+City
ln Output	0.157^{***}	—	—	—
	(0.037)			
$\ln K$	-0.101^{***}	—	—	—
	(0.033)			
$\ln Q$	-0.126^{***}	-0.122^{***}	-0.061^{***}	-0.061^{***}
	(0.010)	(0.009)	(0.008)	(0.008)
$\ln n^B$	0.045	0.132^{***}	0.106^{**}	0.106^{**}
	(0.060)	(0.032)	(0.052)	(0.052)
$\ln n^S$	0.138^{*}	-0.008	0.012	0.012
	(0.075)	(0.040)	(0.040)	(0.040)
$\ln q^m$	0.077^{***}	0.033^{**}	-0.078^{***}	-0.078^{***}
	(0.021)	(0.016)	(0.022)	(0.022)
$\ln \bar{p}^B$	0.503^{***}	0.403^{***}	-0.008	-0.008
	(0.023)	(0.019)	(0.021)	(0.021)
$\ln \bar{p}^S$	0.242^{***}	0.219^{***}	0.047^{***}	0.047^{***}
	(0.022)	(0.019)	(0.015)	(0.015)
Cons	-0.938^{*}	1.385^{***}	4.664^{***}	4.664^{***}
	(0.503)	(0.241)	(0.363)	(0.364)
Firm FE	No	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
City FE	No	No	No	Yes
N	28,967	28,636	28,287	28,287
Adj. R^2	0.332	0.472	0.597	0.595

1.5 T5: Full Specification

The full specification combines all three variable groups:

$$\ln p_{fjct} = \boldsymbol{\gamma}'\mathbf{x}_{ft} + \boldsymbol{\beta}'\mathbf{s}_{fj} + \boldsymbol{\lambda}'\mathbf{z}_{jct} + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \tag{5}$$

Key findings.

- In OLS, firm size is positive ($+0.181^{***}$) and capital is negative (-0.074^{**}); similarity S_{mj}, C_{mj} are strongly positive in OLS (1.076^{***}, 0.888^{***}) and attenuate to marginal significance under Product FE (0.197^{*}, 0.258^{*}).
- The buy-side price drops to zero under full FE (-0.011) while the sell-side price stays positive ($\ln \bar{p}^S = 0.047^{***}$).
- Buyer count (0.102^{**}), market volume (-0.082^{***}), and production quantity (-0.068^{***}) all retain explanatory content under full FE.

Table 8: T5: Full Specification

	(1) OLS	(2) +Firm FE	(3) +Firm+Prod	(4) +Firm+Prod+City
$\ln \text{Output}$	0.181^{***}	—	—	—
	(0.033)			
$\ln K$	-0.074^{**}	—	—	—
	(0.029)			
$\ln Q$	-0.158^{***}	-0.138^{***}	-0.068^{***}	-0.068^{***}
	(0.010)	(0.009)	(0.008)	(0.008)
S_{mj}	1.076^{***}	0.386^{***}	0.197^{*}	0.197^{*}
	(0.185)	(0.115)	(0.107)	(0.108)
C_{mj}	0.888^{***}	0.795^{***}	0.258^{*}	0.258^{*}
	(0.229)	(0.149)	(0.139)	(0.140)
$\ln n^B$	0.092	0.135^{***}	0.102^{**}	0.102^{**}
	(0.059)	(0.031)	(0.052)	(0.052)
$\ln n^S$	0.107	-0.003	0.015	0.015
	(0.074)	(0.041)	(0.040)	(0.040)
$\ln q^m$	0.048^{**}	0.025	-0.082^{***}	-0.082^{***}
	(0.020)	(0.016)	(0.022)	(0.022)
$\ln \bar{p}^B$	0.467^{***}	0.393^{***}	-0.011	-0.011
	(0.022)	(0.019)	(0.021)	(0.021)
$\ln \bar{p}^S$	0.228^{***}	0.215^{***}	0.047^{***}	0.047^{***}
	(0.021)	(0.019)	(0.015)	(0.015)
Cons	-1.602^{***}	1.413^{***}	4.712^{***}	4.712^{***}
	(0.491)	(0.236)	(0.363)	(0.364)
Firm FE	No	Yes	Yes	Yes
Product FE	No	No	Yes	Yes
City FE	No	No	No	Yes
N	28,957	28,626	28,277	28,277
Adj. R^2	0.352	0.476	0.598	0.595

1.6 T6: Interactions — Firm Size × Market Conditions

To explore heterogeneous price-setting, all specifications include firm, product, and city FE; $\ln \text{Output}$ enters only through its interactions with the market variables. The production quantity $\ln Q_{fjt}$ is always included. The equation is

$$\ln p_{fjct} = \gamma_3 \ln Q_{fjt} + \boldsymbol{\lambda}'\mathbf{z}_{jct} + \sum_k \mu_k \big(\ln \text{Output}_{ft} \times z^k_{jct}\big) + \delta_f + \delta_j + \delta_c + \varepsilon_{fjct}. \tag{6}$$

The interaction terms vary at the firm×product level and remain identified under all three FE.

Table 9: Variable Definitions — Table T6 (interaction terms)

Variable	Definition
$\ln \text{Output} \times \ln n^B$	Size × buyer-market thickness.
$\ln \text{Output} \times \ln n^S$	Size × seller competition.
$\ln \text{Output} \times \ln \bar{p}^B$	Size × buy-side market price level.
$\ln \text{Output} \times \ln \bar{p}^S$	Size × sell-side market price level.
$\ln \text{Output} \times \ln q^m$	Size × market volume.

Key findings.

- All columns are full-FE specifications; $\ln \text{Output}$ enters only through the interaction terms, so its main effect is not separately estimated.
- The single-interaction columns reveal size heterogeneity: larger firms pay lower prices in markets with more buyers ($\hat{\mu}_1 = -0.019^{***}$) and more sellers ($\hat{\mu}_2 = -0.023^{***}$), and are more sensitive to both the buy-side and sell-side market prices ($+0.011^{***}$ each; the corresponding main effects turn negative when their interaction is added).
- In the full specification (column 6), multicollinearity renders most single interactions insignificant; only the size–volume interaction (-0.011^{***}) remains robust. $\ln Q$ is robustly negative across all columns.

Table 10: T6: Interactions — Firm Size × Market Conditions

	(1) Baseline	(2) +Size×nB	(3) +Size×nS	(4) +Size×$\bar{p}^B$	(5) +Size×$\bar{p}^S$	(6) All
$\ln Q$	-0.047^{***}	-0.047^{***}	-0.047^{***}	-0.047^{***}	-0.047^{***}	-0.046^{***}
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$\ln n^B$	0.133^{***}	0.516^{***}	0.138^{***}	0.138^{***}	0.138^{***}	0.009
	(0.040)	(0.101)	(0.040)	(0.040)	(0.040)	(0.148)
$\ln n^S$	-0.003	-0.010	0.457^{***}	-0.005	-0.006	0.211
	(0.035)	(0.035)	(0.115)	(0.034)	(0.035)	(0.156)
$\ln q^m$	-0.083^{***}	-0.084^{***}	-0.085^{***}	-0.087^{***}	-0.085^{***}	0.134^{*}
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.072)
$\ln \bar{p}^B$	-0.018	-0.018	-0.019	-0.236^{***}	-0.017	0.083
	(0.017)	(0.017)	(0.017)	(0.073)	(0.017)	(0.087)
$\ln \bar{p}^S$	0.052^{***}	0.052^{***}	0.052^{***}	0.052^{***}	-0.161^{**}	-0.075
	(0.011)	(0.011)	(0.011)	(0.011)	(0.064)	(0.074)
Size×nB		-0.019^{***}				0.007
		(0.005)				(0.007)
Size×nS			-0.023^{***}			-0.011
			(0.006)			(0.008)
Size×$\bar{p}^B$				0.011^{***}		-0.005
				(0.004)		(0.004)
Size×$\bar{p}^S$					0.011^{***}	0.006
					(0.003)	(0.004)
Size×qm						-0.011^{***}
						(0.004)
Cons	4.749^{***}	4.702^{***}	4.726^{***}	4.786^{***}	4.773^{***}	4.744^{***}
	(0.298)	(0.291)	(0.290)	(0.299)	(0.299)	(0.290)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
N	45,669	45,669	45,669	45,669	45,669	45,669
Adj. R^2	0.605	0.605	0.605	0.605	0.605	0.606

2 Results versus Model Predictions

This section confronts the estimates with the model's predictions, separating the results that match expectations from those that diverge, and flags the identification issue most worth addressing.

In line with predictions

- **Quantity discount.** The production quantity $\ln Q$ is robustly negative across all specifications (OLS -0.177^{***}, full FE -0.047^{***}), consistent with the expectation that larger purchase volumes command lower unit prices.
- **Size $\times$ market-thickness interactions are negative.** Larger firms pay lower prices in markets with more buyers ($\mu_1 = -0.019^{***}$), more sellers ($\mu_2 = -0.023^{***}$), and larger total volume ($\text{Size}\times q^m = -0.011^{***}$), matching the prediction that size confers bargaining power.
- **Sell-side price carries independent information.** The sell-side leave-one-out price $\ln \bar{p}^S$ stays significantly positive under full FE (0.052^{***}), consistent with same-side market price levels passing through to a firm's purchase price.

Diverging from predictions

- **Firm-size main effect has the wrong sign (the key divergence).** The model predicts that larger firms, with stronger bargaining power, pay lower prices; but in OLS $\ln \text{Output}$ is significantly positive ($+0.234^{***}$). Crucially, the size$\times$market interactions are *negative*, meaning within a given market larger firms do bargain prices down — only the size *main effect* (across-market comparison) is positive. The positive sign therefore more likely reflects a quality/positioning selection effect (larger firms source higher-quality inputs) rather than bargaining power, so $\ln \text{Output}$ should not be read as a proxy for bargaining power.
- **Product similarity has the wrong sign.** Theory predicts that the more similar the outsourced product is to the firm's own (technological proximity, less information asymmetry, even the option to self-produce), the lower the purchase price; yet S_{mj} and C_{mj} are strongly positive in OLS ($+1.491^{***}$, $+1.481^{***}$) and decay to barely significant once Product FE are added (0.186^{*}, 0.263^{*}).
- **Seller count is insignificant.** The model predicts that more sellers (greater supply-side competition) lower prices, yet $\ln n^S$ is insignificant in the main regressions, showing no such competition effect.
- **Buyer count is positive.** $\ln n^B$ is robustly positive across specifications (0.115^{***}), opposite to the naive prediction that more demand weakens bargaining; it would need reinterpreting as a demand-congestion or market-buoyancy effect.
- **Market volume is sign-unstable.** $\ln q^m$ is positive in OLS ($+0.077^{***}$) but turns significantly negative once Product FE are added (-0.100^{***}); the sign flips with the specification, indicating that what it captures differs before and after controlling for product mix.

Data caveat. One important caveat: the price p_{fjct} is computed as invoice value divided by quantity, but the *units of quantity are not yet harmonized* (different firms and products may report in kilograms, tonnes, pieces, boxes, etc.). Inconsistent units make unit prices within the same 9-digit product not directly comparable and inject noise into the coefficients above (especially the sign and magnitude of the level effects). A follow-up should harmonize the units—or recompute unit prices at a finer product×unit level—and re-check, to confirm that the divergences in this section are driven by economic mechanisms rather than by a units-of-measure artifact.

3 SQL Data Extraction Logic

Background

Raw data originate from the national VAT invoice database for 2017, stored in a relational database. Five aggregated views were extracted and saved as flat files for downstream Python processing. Table 11 summarizes each view.

Table 11: Extracted Data Views from the VAT Database

View name	Unit of observation	Raw / Clean rows	Contents
`firm_buy`	Buyer firm × product code	473,268 / 425,713	Purchase amount and quantity; 2,731 distinct products.
`firm_sell`	Seller firm × product code	101,844 / 89,584	Sales amount and quantity; 2,542 distinct products. Sales quantity feeds $\ln Q_{fjt}$ and the sell-side leave-one-out price.
`firm_city`	Firm ID × prefecture city	3,410 / 3,410	City assignment for each sample firm. Used to merge city into purchase/sale files which contain no city field.
`city_buy`	Buyer prefecture × product code	2,104,901 / 1,393,329	Universe-level buyer count, amount, and quantity. Used to construct n^B, q^m, and the leave-one-out $\bar{p}^B$.
`city_sell`	Seller prefecture × product code	1,540,065 / 954,222	Universe-level seller count, amount, and quantity. Used to construct n^S and the leave-one-out $\bar{p}^S$.

Core Python Processing Steps

1. **Product code standardization.** Raw codes are right-padded to 19 digits; the first 9 digits are retained as the standardized product code. Only the 2,778 valid codes from the national product catalog are kept.

2. **Outsourcing identification.** A firm-product pair is classified as an outsourcing transaction if and only if the firm appears in *both* `firm_buy` and `firm_sell` for the same product code (i.e., the firm both purchases and sells the same product), with min(purchase value, sales value) > 0.

3. **City merge.** Because the purchase and sale files contain no geographic identifier, `firm_city` is joined to both files on firm ID to assign each transaction to a prefecture.

4. **Market condition construction.** Universe views `city_buy` and `city_sell` are aggregated at the product×city level to produce n^B_{jct}, n^S_{jct}, q^m_{jct}, $\bar{p}^B_{jct}$, and $\bar{p}^S_{jct}$. Both market average prices are constructed on a leave-one-out basis: each firm's own value and quantity (on the buy and sell side respectively) are subtracted from the city–product totals before taking the ratio, so the controls do not mechanically contain the firm's own price (firms that are the sole buyer/seller are set to missing). These aggregates are *not* restricted to sample firms, ensuring that the market measures capture the full competitive environment.

5. **Intermediary removal.** Firms flagged as intermediaries (`is_intermediary == 1`; outsourcing ratio > 90%) are dropped from the 2,108-firm outsourcing panel. This removes 233 firms, leaving **1,875 firms** in the final regression panel (`reg_panel.dta`).

Sample Summary

Table 12: Sample Summary Statistics

Statistic	Value
Original sample firms	3,410
Firms with valid buy-side product codes (bianma)	3,376
Outsourcing firms (`invoice_panel.dta`)	2,108
(outsourcing = same product bought & sold)	
Regression panel firms (`reg_panel.dta`)	1,875
(after dropping 233 intermediaries)	
Avg. outsourcing products per firm (mean / median)	25.0 / 3
Outsourcing product types (across all firms)	2,143
Regression observations (T3/T6, full FE)	≈45,669
Cities covered	262

4 Bridge Data Source and Matching Rate

Purpose

The VAT invoice database contains transaction-level information but no firm financial data. To include $\ln K_{ft}$ (log total assets) as a regressor in Tables T1, T2, T4, and T5, firm identifiers are bridged to a separate tax-filing dataset that records balance-sheet information.

Bridge Path

$$\underbrace{\text{Firm ID (VAT, \texttt{cid})}}_{\text{VAT database}} \longrightarrow \underbrace{\text{Tax-filing matched file}}_{\text{bridge key}} \longrightarrow \underbrace{K_{ft} \text{ (total assets, double precision)}}_{\text{tax filing data}}$$

Matching Rate

Table 13: Bridge Matching Rate

Item	Value
Regression panel firms (post intermediary removal)	1,875
Successfully matched firms (with $\ln K$)	≈1,168
Firm-level matching rate	62.3%
Regression observations: T1/T2/T4/T5 (require $\ln K$)	≈29,000
Regression observations: T3/T6 (no $\ln K$, full FE)	≈45,669
Observation loss from capital requirement	≈37%